

Thi Vu

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RESEARCH INTERESTS

I am a Predoctoral Research Resident at Qualcomm AI Research Residency Program, a prestigious program that prepares Vietnam's top undergraduates to pursue PhD in AI. My research interests include **Multimodal AI, Speech Foundation Models, Human-AI Interaction, AI for Creativity**.

EDUCATION

University of Engineering and Technology, Vietnam National University, 2020–2024
Hanoi (UET, VNU)
Bachelor in Computer Science GPA: 3.63/4.0

PUBLICATIONS

Thi Vu, Linh The Nguyen, and Dat Quoc Nguyen. “Zero-Shot Text-to-Speech for Vietnamese”. In: *Proceedings of ACL*. Metareview 5/5. 2025. URL: <https://arxiv.org/pdf/2506.01322>.

Thi Vu, Linh The Nguyen, and Dat Quoc Nguyen. “Vietnamese Automatic Speech Recognition: A Re-visit”. Under review at ACL Rolling Review. Oct 2025.

WORK EXPERIENCE

Qualcomm AI Research | *Research Resident* Hanoi, Vietnam
Advisor: Dr. Dat Quoc Nguyen Apr 2025–Present

- Led development of the first Vietnamese ASR dataset that offers (1) large scale, (2) full orthographic formatting (punctuation/capitalization), and (3) word-level timestamps for **structured, human-readable outputs**. Authored a first-author paper submitted to **ARR Oct 2025** (targeted for ACL 2026).
- Evaluated how accent coverage, timestamp ratio, and punctuation/capitalization conditioning influence transcription and timestamp alignment quality.
- Applied Rotation Program: developed retrieval QA for on-device chat messages with compact embedding indexes, hybrid keyword+semantic search, reranking, and latency-aware caching.

VinAI Research | *Research Resident* Hanoi, Vietnam
Advisor: Dr. Dat Quoc Nguyen Sep 2023–Mar 2025

- Built and released **PhoAudiobook**, a 1500h dataset addressing the **resource gap** for Vietnamese speech; designed a scalable, **language-agnostic data pipeline** for processing 23k+ hours of audio. Downloaded 2000+ times per month. [Hugging Face](#).
- Benchmarked zero-shot TTS models using objective metrics and **human evaluations**; first-author paper accepted to **ACL 2025 Main Track** (metareview 5/5).

Institute for Artificial Intelligence | *Research Assistant* UET, VNU
Advisor: Dr. Tran Quoc Long 2022–2023

- Led a 6-student team on ship segmentation in satellite imagery: built baseline, conducted error analysis, iterated with data/model/augmentation improvements, and deployment. The project successfully secured funding from Vietnam National University.
- Organized weekly seminars to mentor junior students on machine learning fundamentals and research methodologies.

PROJECTS

Vietnamese Braille Keyboard (VBK)

[GitHub](#) | [Demo](#)

An inclusive accessibility tool empowering visually impaired users to type efficiently on smartphones. Features a novel 6-dot touch input interface with interactive voice guidance.

BVoice

[GitHub](#)

Bridged the communication gap for the deaf community by deploying an on-device sign language recognition model using MediaPipe for real-time video-to-text translation.

PhoTextNormalization

[Hugging Face](#)

Fine-tuned mBART-large-50 model on Vietnamese written to spoken text pairs for robust text normalization (e.g., numbers/dates → spoken form).

Billboard Visual Data Analysis

[GitHub](#)

Exploratory analysis of music trends (1960–2022) combining NLP on lyrics with Spotify audio features to visualize the evolution of musical attributes over decades.

Stress Reducer

[GitHub](#) | [Demo](#)

Developed a human-centered health app implementing neuroscience-backed breathing protocols (NSDR) with guided audio-visual cues for rapid stress reduction.

AWARDS & SCHOLARSHIPS

- Outstanding Resident Award in Research 2025 (out of 35 residents), Qualcomm AI Research Vietnam.
- Third Prize (top 3/145) at SOICT hackathon 2023 organized by HUST and Samsung Vietnam.
- Pony Chung Scholarship (by Hyundai Motor Group) — 4 recipients at university level.
- Vietcombank Scholarship — 15 recipients at university level.
- Dean's List (top GPA in program of 300 students in each semester), awarded 3 times.

TECHNICAL SKILLS

- Machine Learning: PyTorch, Transformers, PyTorch Lightning, Hydra, TensorBoard, Weights & Biases, Triton Inference Server
- Audio Processing: torchaudio, librosa
- Mobile Development: Android (Android Studio, Android SDK), MediaPipe AAR, Android TTS.
- Languages: Python, Java, C++
- Tools: Git, Linux, SLURM, LaTeX